Circular Motion Kinematics

Synopsis: I summarize the rotational kinematics equations. Note that rotational kinematics is just like linear kinematics. Instead of linear displacement, you have an angular displacement since the motion has circular symmetry, so the distance from the center of rotation is fixed, and only the angle subtended by the motion changes while rotating. There are equations for angular velocity and angular acceleration that look just like the equations for velocity and acceleration in linear kinematics.

The angular displacement gives the arc length or the actual distance covered, and the angular velocity gives the tangential velocity for a given radius of circular trajectory (see below). Angular displacement is θ, and the arc length is $s = r\theta$, provided θ is in radians. That proviso is why we bother with radians at all: in degrees the same statement needs a conversion factor bolted onto it, and you would carry that factor through every line of every derivation that follows. Angular velocity is $\omega = \Delta\theta/\Delta t$ in rad/s, and the tangential velocity it gives you at radius r is $v_t = r\omega$. Notice that ω has no r in it and v_t does: every point on a rigid turning object has the same angular velocity, and a different speed.

We close out with equations for centripetal acceleration, which connect the force required to keep a body in circular motion to its angular velocity. The velocity of an object in circular motion is perpendicular to the radius always – that is, tangential to the circle. The net force on it is parallel to the radius and points toward the center of the circle, and center-pointing is what the word centripetal means. Put those two together and v and F are perpendicular to each other.

Everything in this Note is uniform circular motion only. When the tangential speed is constant and r is constant, ω is constant, so the centripetal acceleration is constant, so the centripetal force is constant, and the trajectory is a uniform circle of radius r traced at a fixed spin rate.

Performance Expectations: Apply rotational kinematics equations to solve circular and centripetal force problems.

For PHY 151 — with calculus: Define the angular variables as derivatives, $\omega = d\theta/dt$ and $\alpha = d\omega/dt$, recover the kinematics equations by integration, and get $a = -\omega^2 r$ by differentiating the position vector twice.

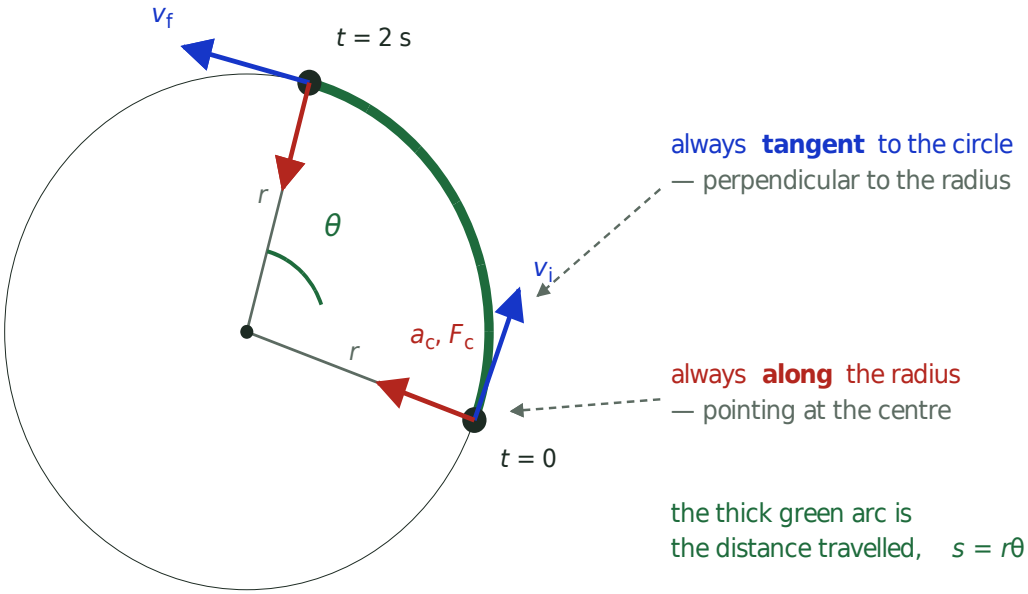
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Before you read on – predict. Two children sit on the same merry-go-round, one near the hub and one out at the rail. Are they moving at the same speed, and are they turning at the same rate? Commit to a sentence here, before you watch. You will come back to it at the end.

Same speed? Same rate? One sentence.

Instructions and Questions

1) Angular displacement, and the radian.
Fix a center and a radius and the only thing left to say about a particle on that circle is its angle. Two things to keep straight: the arc is the distance actually travelled, so s = rθ is a real distance in metres; and that formula is true in radians and false in degrees.



Fill in the four names for one full turn, and the arc-length formula with the condition that makes it true:

1 _____ = 1 _____ = 1 _____ = _____° = _____ radians arc length s = _____ true only when θ is in _____ degrees → radians: × _____ radians → degrees: × _____ π rad = _____ °

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Complete the table.

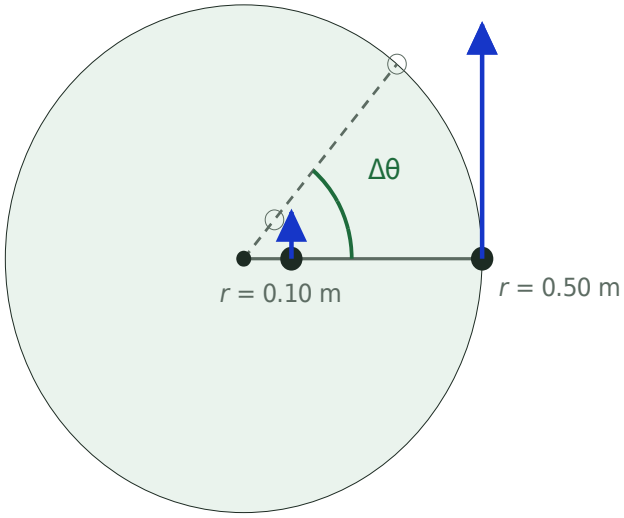
Angle	**in radians**	**in degrees**	**in revolutions**
10°		—	
50°		—	
75°		—	
260°		—	
π/3 rad	—		
0.55π rad	—		
1.8π rad	—		
4π rad	—		

i) A sector of radius 2.0 m subtends 75°. Find the arc length twice – once as $d = 2\pi r(\theta°/360)$, once as $s = r\theta$ – and say why they agree.

2) Angular velocity, and the tangential speed it gives you.
Every rotating machine in every problem is quoted in rev/min, and every equation below wants rad/s, so the first line of the working is almost always the same conversion. Write the three you need:

ω = _____ units: _____ v_t = _____
rev/min → rad/s: ω = (rev/min) × _____ ≈ (rev/min) × _____

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Both riders sit on the same spoke of the same merry-go-round.

Δθ is the angle the spoke sweeps; the open circles are where each rider has got to after that same interval.

Two riders on one spoke of the same merry-go-round. Radii and velocity arrows are both drawn to scale.

i) Microwave ovens rotate at a rate of about 7 rev/min. (A) What is this in revolutions per second? (B) What is the angular velocity in radians per second?

ii) A merry-go-round makes 2.5 rotations in 20 s. (A) What is its angular velocity? (B) What is the speed of a rider sitting 0.50 m from the center? (C) And of a rider sitting 0.10 m from the center?

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iii) In lacrosse, a ball is thrown from a net on the end of a stick by rotating the stick and forearm about the elbow. If the angular velocity of the ball about the elbow joint is 30.0 rad/s and the ball is 1.2 m from the elbow joint, what is the velocity of the ball?

iv) From your answers to the merry-go-round: which of ω and v_t depends on where you sit, and which does not? One sentence.

3) The rotational kinematics equations.
Define α = Δω/Δt, the angular acceleration, in rad/s^2. Now every constant-acceleration equation from Module 1 has a twin with θ in place of d – not an analogy, the same derivation in different letters. If you know the left column you already know the right one. This is the page you will turn back to for the rest of the module, so make it the neat one.

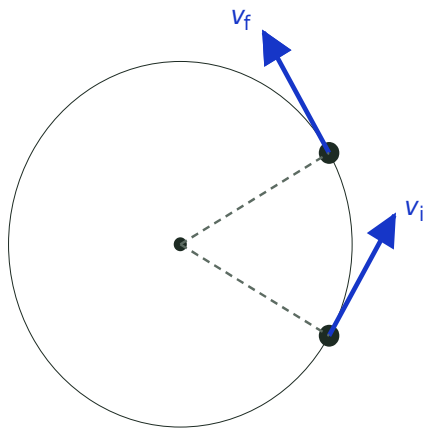
Linear Kinematics	**Rotational kinematics**
$d = d_0 + vt + \frac{1}{2}at^2$	
$v = v_0 + at$	
$v^2 = v_0^2 + 2ad$	
arc length	
tangential velocity	
tangential acceleration	
centripetal acceleration	
centripetal force	

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i) One of the bridge rows plays no part anywhere in this module. Which one, and why not?

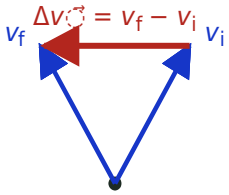
ii) What unit must ω be in for every row to work, and by what factor is an answer wrong if you feed in rev/min instead?

4) Why turning at constant speed is accelerating.
Acceleration is change of velocity, and velocity is a vector. Go round a circle at a perfectly steady speed and the size never changes, but the direction changes continuously, so Δv is not zero. Subtract the two velocity vectors and look at where the difference points.



two instants, a short time apart

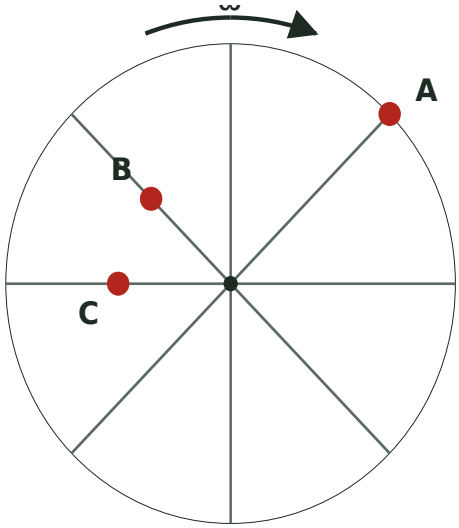
redraw both from one tail



i) Redraw the construction yourself, then say in one or two sentences why the object is accelerating and which way the acceleration points.

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A rigid wheel of radius r rotates about an axis through the center and perpendicular to the plane of the wheel. Point A is on the rim; points B and C are on two different spokes, each at a distance r/2 from the center. The next three questions are about one interval Δt of that rotation.



A is on the rim, at radius *r*.

B and C are on two different spokes, each at *r*/2 from the centre.

The wheel is rigid and turns as one.

ii) Which of the following is true regarding the **angular velocity**? (A) All three points have the same angular velocity, as all three have the same angular displacement θ in the same time interval Δt. (B) Points B and C have half the angular velocity of point A, as they are at half the distance from the center. (C) Points B and C have twice the angular velocity of point A, as they are at half the distance from the center.

iii) Which of the following is true regarding the **tangential speed**? (A) All three points have the same tangential speed, as all three have the same angular displacement θ in the same time interval Δt. (B) Point A has a greater tangential speed as it is further from the center of the wheel. (C) Points B and C have a greater tangential speed as they are closer to the center.

iv) Which of the following is true regarding the **centripetal acceleration**? (A) All three points have the same centripetal acceleration, as all three have the same angular displacement θ in the same time interval Δt. (B) Points B and C have half the centripetal acceleration of point A, as they are at half the distance from the center. (C) Points B and C have twice the centripetal acceleration of point A, as they are at half the distance from the center.

Working for all three – which equation decides each one?

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5) Centripetal force.
One warning about the word before you use it: centripetal is a job description, not a kind of force. You never draw an arrow labelled "centripetal force" on a free-body diagram. You draw the forces that are really there – tension, friction, gravity, a normal force – and then notice that their sum points at the center. Whichever force is doing that job is the centripetal force in that problem.

$a_{centripetal}$ = _____ = _____

$F_{centripetal}$ = _____ = _____ = _____

Forces that can do the job: _____

i) An ordinary workshop grindstone has a radius of 8.50 cm and rotates at 7500 rev/min. (A) Calculate the magnitude of the centripetal acceleration at its edge in metres per second squared, and convert it to multiples of g. (B) What is the linear speed in m/s of a point on its edge?

ii) A 23.0 kg child is riding a playground merry-go-round that is rotating at 40.0 rev/min. What centripetal force (in N) must she exert to stay on if she is 2.00 m from its center?

iii) What centripetal force (in N) does she need to stay on the same merry-go-round, at the same rotation rate, if she is 6.00 m from its center? Compare each force with her own weight, and say what that means for someone trying to hold on at the rail.

6) For PHY 151 — with calculus.
The ratios above are averages over Δt. The instantaneous quantities are the derivatives, exactly as in linear kinematics: $\omega = d\theta/dt$ and $\alpha = d\omega/dt = d^2\theta/dt^2$. Written that way, the three rotational kinematics

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equations are not an analogy you have to trust – they are what you get by integrating α twice, which is the Module 1 calculation in different letters.

i) Integrate α once and then twice to recover $\omega = \omega_0 + \alpha t$ and $\theta = \theta_0 + \omega_0 t + \tfrac{1}{2}\alpha t^2$, and say exactly where the assumption "α constant" is used.

ii) Write $r(t) = r(\cos \omega t)\, i + r(\sin \omega t)\, j$ for uniform circular motion and differentiate twice. What does the result say about the direction of the acceleration, with no limit argument and no vector triangle?

iii) Verify that dr/dt is perpendicular to r at every instant and that its magnitude is rω.

iv) Repeat the differentiation with ω a function of time. A second term appears – identify it in the comparison table.

Back to your prediction. Go back to the box on the first page. Was it the same rate, the same speed, both, or neither? Correct it here in one sentence, using ω and v_t by name.

Check yourself

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Four to do once you have filled the notes in. Work them here, then check them against the interactive version at the bottom of the note page.

1) A wheel turns at 45.0 rev/min. What is its angular velocity in rad/s?

2) Two riders sit on the same turning merry-go-round, one at 1.0 m from the hub and one at 3.0 m. Which statement is true?
   i) Same tangential speed, and the outer rider has one third the angular velocity
   ii) Same angular velocity, and the outer rider has three times the tangential speed
   iii) Same angular velocity and the same tangential speed – it is one rigid object
   iv) The outer rider has three times the angular velocity and three times the tangential speed

Why?

3) An object moves in a circle at constant speed. The net force on it
   i) is zero, because the speed is not changing
   ii) points along the direction of motion
   iii) points at the center, perpendicular to the velocity
   iv) points outward, away from the center

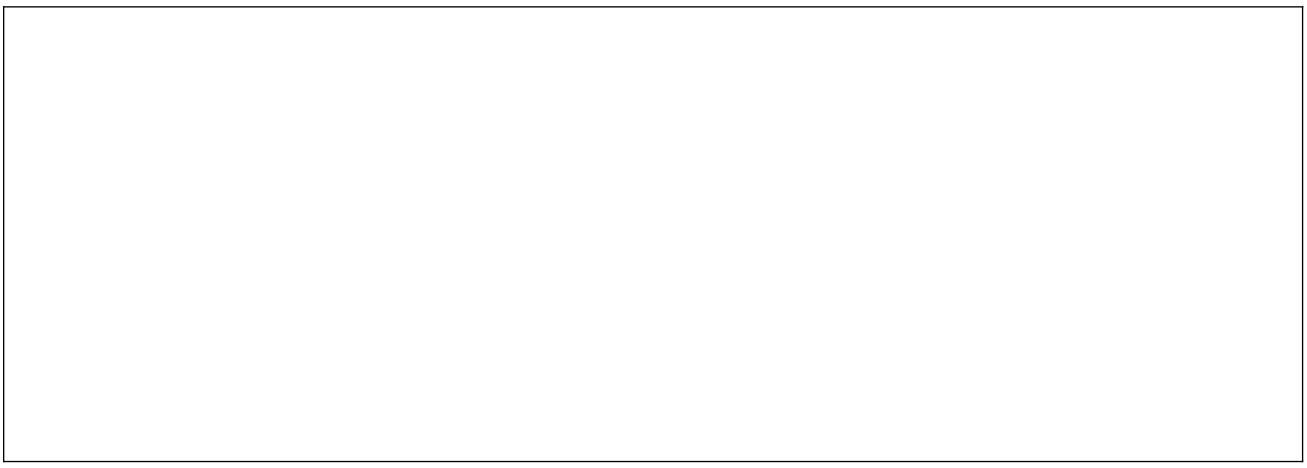
Why?

4) A 0.50 kg puck is swung in a horizontal circle on a frictionless table, on a string 1.20 m long, at an angular velocity of 3.0 rad/s. What is the tension in the string, in N?

Your notes

Space for the sketches, the derivations you want to redo yourself, and anything from the video that did not fit above. Everything on this handout is uniform circular motion; the later notes in this module turn α back on and take the same angular variables into rotational dynamics – torque, moment of inertia and angular momentum.

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